

Developing changepoint methodology motivated by climate applications

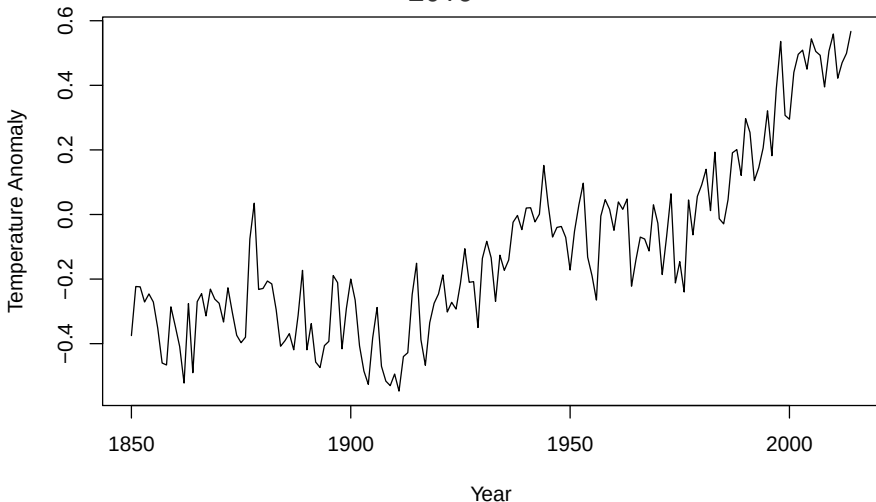
Rebecca Killick(r.killick@lancs.ac.uk)

Joint work with Claudie Beaulieu (Uni California Santa Cruz)

Parts with: Adelia Johnson, Colin Gallagher, Robert Lund, Xueheng Shi
IDWSDS 2026

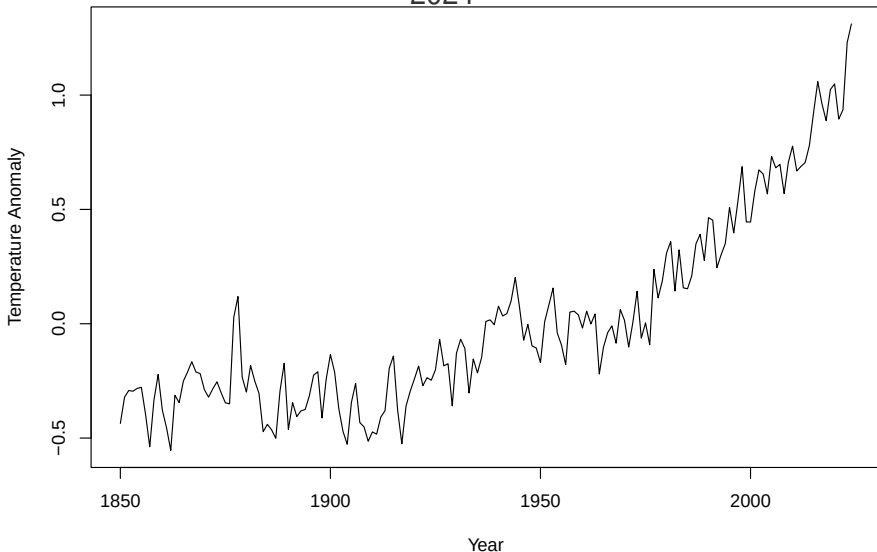
Where my interest began

2016



Global Mean Temp

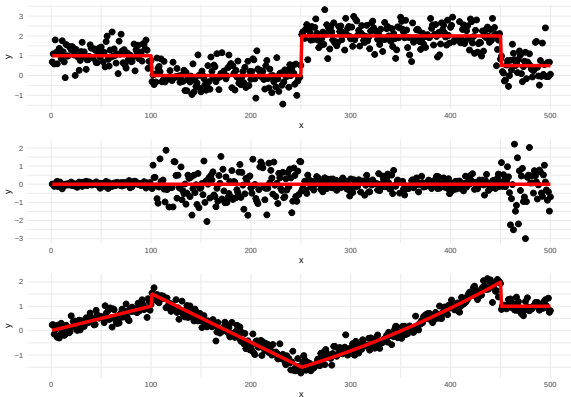
2024



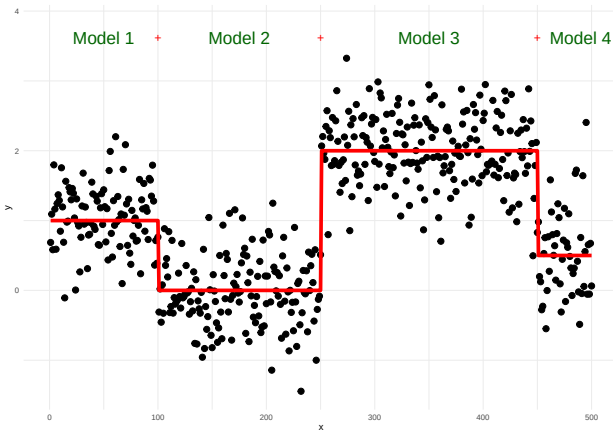
What are changepoints?

For data $y_1, \dots, y_n$, if a changepoint exists at τ , then $y_1, \dots, y_\tau$ differ from $y_{\tau+1}, \dots, y_n$ in some way.

There are many different types of change.

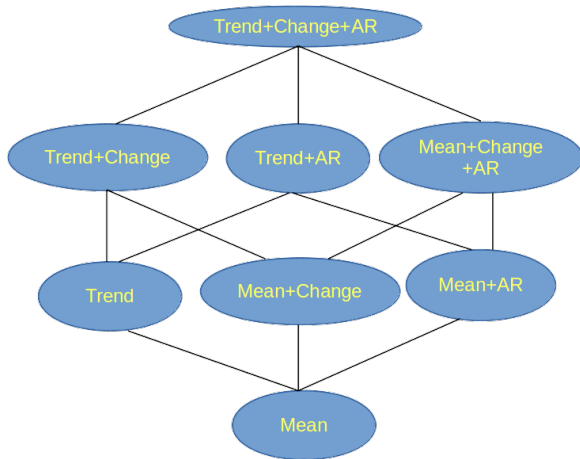


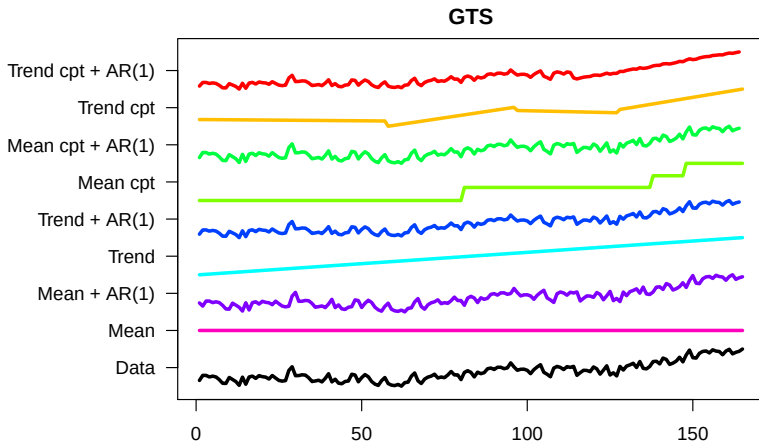
Problem



- How many changes?
- Where are the changes? 2^{n-1} possible solutions!

EnvCpt: select the most parsimonious but accurate model for the data. Simple to extend with other types of models.



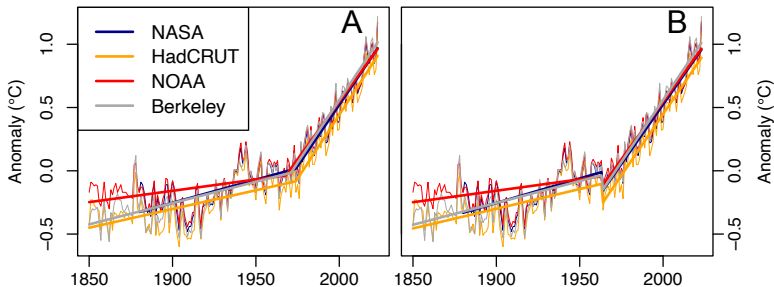


$$Y_t = \theta_i + \frac{\theta_{i+1} - \theta_i}{\tau_{i+1} - \tau_i}(t - \tau_i) + Z_t$$

but Z_t is AR(p), $Z_t = \phi_1 Z_{t-1} + \phi_2 Z_{t-2} + \dots + \phi_p Z_{t-p} + \epsilon_t$.

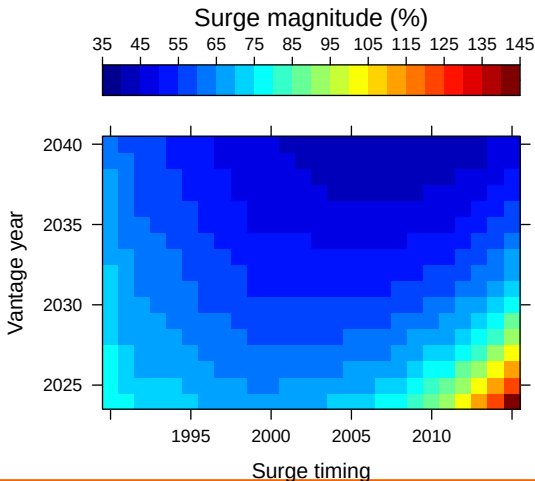
Challenge:

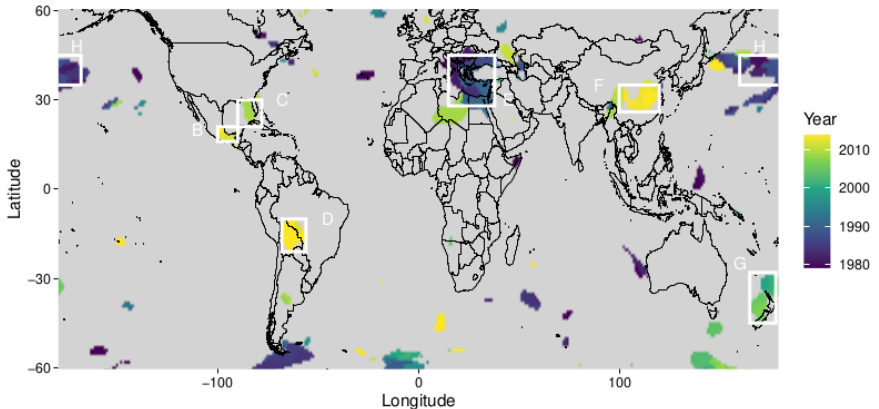
- Definition of join-pin
- Fitting AR parameter across segments:
 - Fixed we use EM algorithm
 - Varying we can embed in PELT

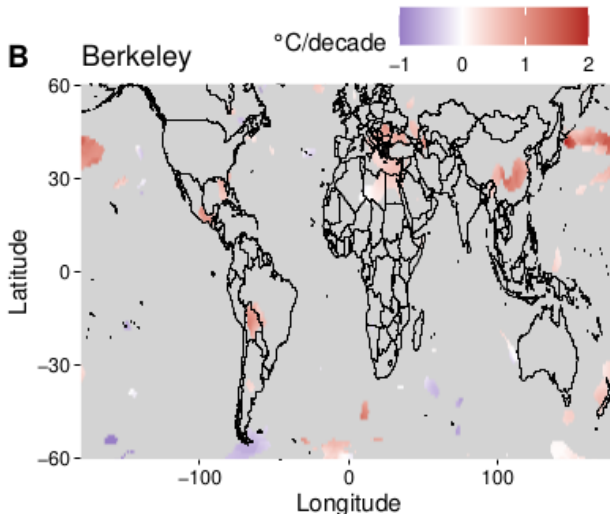


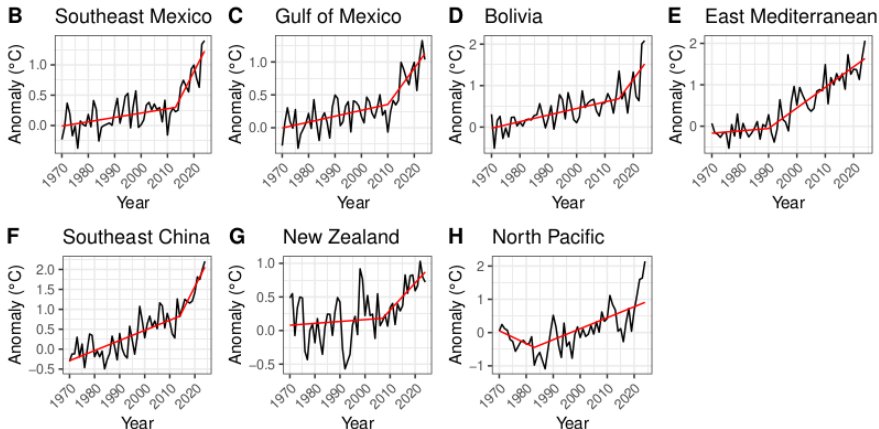
So what?

What would we need to see (in the statistically preferred model).









- Not accounting for autocorrelation can lead to spurious changes!
- Detecting and documenting changepoints improves analyses . . .
- . . . and can help answer pertinent questions in different domains

- Extending to more complex model structures is interesting statistically
- Extending to include covariates is interesting climatically

- I enjoy working with different disciplines . . .
- . . . as often it sparks my next research challenge.

Presentations available at: <https://github.com/rkillick/Presentations>

Preprints of all available at: www.lancs.ac.uk/~killick/pub.html

Choice of Model (Beaulieu, Killick 2018):
<https://doi.org/10.1175/JCLI-D-17-0863.1>

Warming Surge (Beaulieu et al., 2024):
<https://doi.org/10.1038/s43247-024-01711-1>

Regional warming surge (Beaulieu et al., 2026):
<https://doi.org/10.1038/s41467-026-77514-z>